

Writing Recursive Rules for Arithmetic Sequences

Finding the first term and the common difference, writing recursive rules, and extending sequences

Directions

- A recursive rule for an arithmetic sequence has two parts: a **starting value** a_1 and a **step** $a_n = a_{n-1} + d$ for $n \geq 2$. You need both.
 - None of these problems hands you a_1 outright — find the step first, then walk back to term 1.
 - The step between term j and term k is spread over $k - j$ jumps, so $d = \frac{a_k - a_j}{k - j}$.
 - Show the arithmetic that gets you each value; a bare number earns no credit.
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1. An arithmetic sequence has second, third and fourth terms 11, 18, 25.

(a) Find the step. $d =$ _____

(b) Find the starting value the recursive rule needs. $a_1 =$ _____

The recursive rule is then $a_n = a_{n-1} + d$ for $n \geq 2$, using your two values.

(c) Use the rule to reach the tenth term. $a_{10} =$ _____

2. In an arithmetic sequence, $a_3 = 20$ and $a_6 = 8$.

(a) Find the step. $d =$ _____

(b) Find the starting value. $a_1 =$ _____

(c) Use the recursive rule to reach the ninth term. $a_9 =$ _____

3. A theatre gains the same number of seats in every row going back. Row 4 holds 18 seats and row 8 holds 30 seats.

(a) How many seats are added per row? $d =$ _____

(b) How many seats are in row 1? $a_1 =$ _____

(c) Using the recursive rule $a_n = a_{n-1} + d$, how many seats are in row 12? $a_{12} =$ _____

4. A sequence is defined by $a_1 = 6$ and $a_n = a_{n-1} + 5$ for $n \geq 2$.

(a) List the next three terms. $a_2 =$ _____ $a_3 =$ _____ $a_4 =$ _____

(b) Find the twentieth term. $a_{20} =$ _____

(c) Explain why the step $a_n = a_{n-1} + 5$ on its own does not pin down the sequence.

5. In an arithmetic sequence, $a_5 = 3$ and $a_9 = 23$.

(a) Find the step. $d =$ _____

(b) Find the starting value. $a_1 =$ _____

(c) Use the recursive rule to reach the fifteenth term. $a_{15} =$ _____

6. A student deposits the same amount into an account every week. After week 3 the balance is \$145; after week 7 it is \$225. Write a_n for the balance in dollars after week n .

(a) How much is deposited each week? $d =$ _____

(b) What was the balance after week 1? $a_1 =$ _____

(c) Using the recursive rule, what is the balance after week 12? $a_{12} =$ _____

7. A sequence is defined by $a_1 = 5$ and $a_n = a_{n-1} + 6$ for $n \geq 2$.

(a) Find the tenth term. $a_{10} =$ _____

(b) Find the sum of the first ten terms. Sum = _____

8. In an arithmetic sequence, $a_2 = -1$ and $a_6 = 15$.

(a) Find the step. $d =$ _____

(b) Find the starting value. $a_1 =$ _____

(c) Use the recursive rule to reach the eleventh term. $a_{11} =$ _____

9. The rungs of a ladder are evenly spaced. Rung 2 sits 34 cm above the ground and rung 5 sits 79 cm above it. Write a_n for the height of rung n in centimetres.

(a) How far apart are consecutive rungs? $d =$ _____

(b) How high is rung 1? $a_1 =$ _____

(c) Using the recursive rule, how high is rung 10?

Answer: _____ cm

10. Two arithmetic sequences start on the same day. Sequence A has $a_1 = 3$ and $a_n = a_{n-1} + 5$; sequence B has $b_1 = 57$ and $b_n = b_{n-1} - 4$.

(a) At which term number are the two sequences equal? $n = \underline{\hspace{2cm}}$

(b) What is that common value? value = $\underline{\hspace{2cm}}$